```python
# Program to add two matrices

rows = int(input("Enter number of rows: "))
cols = int(input("Enter number of columns: "))

print("Enter elements of first matrix:")
A = []
for i in range(rows):
    row = []
    for j in range(cols):
        row.append(int(input(f"A[{i}][{j}]: ")))
    A.append(row)

print("Enter elements of second matrix:")
B = []
for i in range(rows):
    row = []
    for j in range(cols):
        row.append(int(input(f"B[{i}][{j}]: ")))
    B.append(row)

# Matrix addition
C = []
for i in range(rows):
    row = []
    for j in range(cols):
        row.append(A[i][j] + B[i][j])
    C.append(row)

print("Sum of the two matrices:")
for row in C:
    print(row)
```

```
Enter number of rows: 2
Enter number of columns: 2
Enter elements of first matrix:
A[0][0]: 1
A[0][1]: 2
A[1][0]: 3
A[1][1]: 4
Enter elements of second matrix:
B[0][0]: 5
B[0][1]: 1
B[1][0]: 2
B[1][1]: 4
Sum of the two matrices:
[6, 3]
[5, 8]
```